

# Nano Reactor - Capstone Submission Report

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**Course:** AIGC 5604 — Applied AI for Digital Arts

**Assessment:** Capstone Project

**Track:** Individual

**Theme:** Futuristic Products

**Submission date:** August 18, 2026

**Deliverables briefly:**

**Key Visual Poster:**



1\_artwork\_prototype/poster/Nano\_Reactor\_FINAL\_poster.jpeg

**Promotional Video Spot (30s, 1920×1080):**

1\_artwork\_prototype/video/Nano\_Reactor\_FINAL\_spot.mov

**Watch:** <https://youtu.be/n1xZHRJyDmg>



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## 1. Executive Summary

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*Nano Reactor* is a speculative B2B campaign for a fictional clean-energy company. Its product, the **NR-114**, is a compact, "walk-away safe," megawatt-scale nuclear module pitched as on-site power for the data centers driving the AI boom. The campaign pairs a cinematic key-visual poster with a 30-second promotional spot, both produced through a multi-tool generative-AI pipeline under close human art direction, and fully documented for provenance, licensing, and ethics.

The core creative tension of advertising clean energy for AI while using energy-intensive AI to create the advertisement is addressed directly in the reflection rather than avoided. The project's lowest peer-review score, which focused on ethics and greenwashing, is treated as the central subject of the essay.

## 2. Concept & Speculative Vision

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| Element                 | Detail                                                                                                                                                       |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product                 | Nano Reactor <b>NR-114 (PWR-Class Module)</b> — compact nuclear power core                                                                                   |
| Positioning             | <i>"Clean, compact nuclear power — a megawatt-scale plant engineered for the data centers behind the age of AI."</i>                                         |
| Key specs (on artifact) | 3.2 MW continuous · Walk-away Safe · 8-yr core life · Zero operational carbon                                                                                |
| Brand                   | <div> <b>Nano Reactor</b></div> <p>nanoreactor.io · two-orbit atom logo</p> |
| Inspiration (disclosed) | <b>Aalo Atomics</b> (aalo.com) for the compact-reactor concept; a <b>DJI poster</b> for background/layout mood — style references only, never copied         |
| Audience                | Hyperscale data-center operators, AI-infrastructure builders, energy investors, sustainability-focused enterprise buyers                                     |

### 3. Production Pipeline & Toolchain

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The work moved deliberately across tools, each chosen for a specific strength:

|                                                                                                                       |                   |       |                    |
|-----------------------------------------------------------------------------------------------------------------------|-------------------|-------|--------------------|
|  _TODO_CHECKLIST.md                  | Today at 9:36 PM  | 2 KB  | Markdo...text file |
| ▼  1_artwork_prototype               | Today at 10:12 PM | --    | Folder             |
| >  poster                            | Today at 8:20 PM  | --    | Folder             |
| >  supplementary                     | Today at 7:56 PM  | --    | Folder             |
| >  video                             | Today at 9:27 PM  | --    | Folder             |
| ▼  2_data_templates                  | Today at 9:37 PM  | --    | Folder             |
|  ethical_licensing_audit.csv         | Today at 9:37 PM  | 3 KB  | CSV Document       |
|  peer_feedback_prototype6_RAW.csv    | Today at 8:26 PM  | 6 KB  | CSV Document       |
|  peer_review_revision_log.csv        | Today at 8:35 PM  | 2 KB  | CSV Document       |
|  prompt_refinement_log.csv           | Today at 9:35 PM  | 17 KB | CSV Document       |
|  provenance_seed_log.csv             | Today at 9:32 PM  | 10 KB | CSV Document       |
|  source_history_links.md             | Today at 9:37 PM  | 8 KB  | Markdo...text file |
| ▼  3_output_resource                 | Today at 8:20 PM  | --    | Folder             |
| >  audio                             | Today at 7:56 PM  | --    | Folder             |
| >  logo                              | Today at 7:57 PM  | --    | Folder             |
| >  poster_iterations                 | Today at 8:20 PM  | --    | Folder             |
| >  poster_source                     | Today at 8:20 PM  | --    | Folder             |
| >  reference                         | Today at 8:20 PM  | --    | Folder             |
| >  video_clips                       | Today at 8:57 PM  | --    | Folder             |
| ▼  4_notes_reflection                | Today at 9:39 PM  | --    | Folder             |
|  ai_tool_disclosure.md               | Today at 9:39 PM  | 3 KB  | Markdo...text file |
|  creative_brief.md                   | Today at 9:37 PM  | 5 KB  | Markdo...text file |
|  critical_reflection.md              | Today at 9:28 PM  | 8 KB  | Markdo...text file |
|  Nano_Reactor_Portfolio_Report.docx | Today at 9:44 PM  | 9 KB  | Word document      |
|  Nano_Reactor_Portfolio_Report.md  | Today at 9:40 PM  | 9 KB  | Markdo...text file |
|  Nano_Reactor_Portfolio_Report.pdf | Today at 9:45 PM  | 17 KB | PDF Document       |
|  README.md                         | Today at 9:45 PM  | 3 KB  | Markdo...text file |

| Tool            | Model                                    | Role                                                                                           |
|-----------------|------------------------------------------|------------------------------------------------------------------------------------------------|
| Claude Design   | Opus 4.8                                 | Design system (132 tokens, 14 components, 23 cards), two-orbit atom logo, editable HTML poster |
| Google Flow     | Nano Banana 2                            | All poster imagery + hero reactor + atom/video stills                                          |
| Google Flow     | Veo 3.1 Lite / Veo 3.1 Fast / Omni Flash | Image-to-video animation of 22 clips                                                           |
| ElevenLabs      | Eleven Multilingual v2                   | Voiceover (final voice: Kal Jones)                                                             |
| Suno            | v4.5-all                                 | Background music (instrumental)                                                                |
| DaVinci Resolve | —                                        | Human video edit & assembly                                                                    |
| Perplexity      | Best                                     | Prompt-engineering research aid (no deliverable asset generated)                               |

## 4. Prompt Engineering & Iteration

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**28 prompts** are logged and analyzed in

`2_data_templates/prompt_refinement_log.csv`

Highlights of the iteration discipline:

- **Hero reactor** was image-referenced from Aalo ([Aalo.com](https://aalo.com)) and then refined through iterative prompts, including requests to make it more cylindrical and better match the reference. This demonstrates human-directed convergence rather than reliance on a single successful generation.
- **Poster** underwent a 10-step conversational refinement process in Nano Banana 2, including a cinematic DJI-style redesign, specification callout sizing, 4K upscaling, resizing to 1240 × 1754, a peer-driven typo correction, a bolder eyebrow treatment, and a logo swap.
- **Video** used a consistent **image-to-video** workflow: first generating a still image, then animating it. Prompts were heavily art-directed and included negative prompts to control the visual outcome, such as framing cooling as *"calm controlled thermal management rather than danger"* and specifying *"no sparks, no warning signals."*
- **Seed control:** Nano Banana 2 and Veo do not expose seed controls. Consistency was maintained by iteratively editing a single evolving composition, with each change documented rather than leaving the process unspecified.
- **Voiceover** required a dedicated **pronunciation script**, with "NR-114" rewritten as "N. R. one one four" and punctuation added to control pacing and delivery.

Every logged prompt carries a source link; the full step-by-step trail is in

`2_data_templates/source_history_links.md`.

## 5. Provenance & Reproducibility

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2\_data\_templates/provenance\_seed\_log.csv tracks **39 assets**, documenting the filename, tool, model, prompt reference, parameters, date, human edits, final use, and source link for each asset. The two-tier system, consisting of a complete asset inventory, curated prompt analysis, and a full-link appendix, supports reproducibility

## 6. Human Creative Control & Editing

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The AI produced the raw material, but the authorship remained human-directed:

- **Concept, references, and design system:** Developed and structured by the authors.
- **Curation and visual direction:** Selected from dozens of iterations, including the hero's final form, typography, and specification messaging.
- **Editing:** The **DaVinci Resolve** edit was completed manually, with all Veo clips, Suno music, and Kal Jones voiceover imported, cut into beats, sequenced, color-managed, and synchronized into the final 30-second spot.
- **Prompt authorship:** Prompts were treated as authored artifacts, incorporating negative prompting and specific art-direction constraints.

## 7. Peer Review & Evidence-Based Revision

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Presented as the **Phase-1 Alpha Prototype** (poster) at the in-class showcase.

Five reviewers scored five criteria

(2\_data\_templates/peer\_feedback\_prototype6\_RAW.csv):

| Criterion     | Avg (1–5)            |
|---------------|----------------------|
| Concept       | 4.60                 |
| Craft         | 4.60                 |
| Accessibility | 4.10                 |
| Originality   | 4.00                 |
| <b>Ethics</b> | <b>3.60</b> (lowest) |

**Documented evidence-based revision (Individual requires 1):** Reviewer r3 identified distorted and misspelled bottom text, including "N2-114" and "nanozeactor.io." This feedback led to logged prompt **P10**, which instructed the correction of "nanozeactor.io" to "nanoreactor.io," along with an enlargement of the specification text. The before-and-after changes are documented in 2\_data\_templates/peer\_review\_revision\_log.csv, with the final poster correctly displaying **NR-114 / nanoreactor.io**.

## 8. Ethics, Licensing & Critical Reflection

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Full essay: `4_notes_reflection/critical_reflection.md`.

Audit: `2_data_templates/ethical_licensing_audit.csv` (11 items).

Key positions:

- **Copyright:** No source image was copied. Aalo and DJI were used only as disclosed style and concept references, while the final imagery was AI-regenerated.
- **Licensing (honest limit):** The Suno music was created using a free plan that does not grant commercial rights. This is acceptable for the non-commercial academic exercise and is disclosed transparently. A real commercial release would require a paid plan or replacement music. Licensing for other tools is marked **"to verify."**
- **Bias:** The models' default **premium tech-ad aesthetic** emerged during generation, with reviewers noting an "Apple-ad" familiarity. Human curation was used to mitigate this influence, while acknowledging that it was not fully eliminated.
- **Deepfakes:** The voiceover uses a **synthetic stock voice**, not a clone of a real person, avoiding concerns related to likeness or consent.
- **Environment:** The campaign's thesis, that AI's power demand is outpacing its infrastructure, also reflects the project's own computational footprint. Iteration discipline is therefore framed as part of the project's environmental practice.
- **Greenwashing:** Peer feedback regarding unqualified claims such as **"clean," "zero-carbon,"** and **"walk-away safe"** was accepted and addressed. These claims are explicitly presented as speculative or fictional throughout the project.

## 9. Repository Map

|   |                                    |       |                    |
|---|------------------------------------|-------|--------------------|
| ▼ | Capstone                           | --    | Folder             |
|   | README.md                          | 3 KB  | Markdo...text file |
|   | Nano_Reactor_Portfolio_Report.pdf  | 17 KB | PDF Document       |
|   | Nano_Reactor_Portfolio_Report.docx | 9 KB  | Word document      |
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|   | _TODO_CHECKLIST.md                 | 2 KB  | Markdo...text file |
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|   | ai_tool_disclosure.md              | 3 KB  | Markdo...text file |
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| ▼ | 3_output_resource                  | --    | Folder             |
| > | poster_source                      | --    | Folder             |
| > | reference                          | --    | Folder             |
| > | logo                               | --    | Folder             |
| > | poster_iterations                  | --    | Folder             |
| > | audio                              | --    | Folder             |
| > | video_clips                        | --    | Folder             |
| ▼ | 2_data_templates                   | --    | Folder             |
|   | source_history_links.md            | 8 KB  | Markdo...text file |
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|   | peer_feedback_...TOTYPE6_RAW.csv   | 6 KB  | CSV Document       |
| ▼ | 1_artwork_prototype                | --    | Folder             |
| > | supplementary                      | --    | Folder             |
| > | video                              | --    | Folder             |
| > | poster                             | --    | Folder             |

## 10. Assessment Alignment (100 pts)



| Component                               | Pts | Where evidenced                           |
|-----------------------------------------|-----|-------------------------------------------|
| Creative Concept & Final Artifact       | 20  | 1_artwork_prototype/                      |
| Technical & Effective Use of AI         | 15  | 2_data_templates/ +<br>3_output_resource/ |
| Human Editing & Creative Control        | 20  | 3_output_resource/ +<br>DaVinci edit      |
| Process Documentation & Provenance      | 20  | 2_data_templates/ (28 prompts, 39 assets) |
| Ethical Reflection & Critical Awareness | 15  | 4_notes_reflection/                       |
| Prototype, Peer Evaluation & Revisions  | 10  | peer review + traced revision             |

*Appendix — full tool disclosure:* 4\_notes\_reflection/ai\_tool\_disclosure.md.

*All prompt source links:* 2\_data\_templates/source\_history\_links.md.